

CATALYSIS

Selective conversion of syngas to light olefins

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Although considerable progress has been made in direct synthesis gas (syngas) conversion to light olefins ($C_2=C_4$) via Fischer-Tropsch synthesis (FTS), the wide product distribution remains a challenge, with a theoretical limit of only 58% for C_2-C_4 hydrocarbons. We present a process that reaches $C_2=C_4$ selectivity as high as 80% and C_2-C_4 94% at carbon monoxide (CO) conversion of 17%. This is enabled by a bifunctional catalyst affording two types of active sites with complementary properties. The partially reduced oxide surface ($ZnCrO_x$) activates CO and H_2 , and C-C coupling is subsequently manipulated within the confined acidic pores of zeolites. No obvious deactivation is observed within 110 hours. Furthermore, this composite catalyst and the process may allow use of coal- and biomass-derived syngas with a low H_2/CO ratio.

Fischer-Tropsch synthesis (FTS) has played an important role as a gas-to-liquid technology, producing synthetic lubricants and synthetic fuels from coal, natural gas, or biomass since it was developed almost a century ago. It is also the only effective technology to date for direct conversion of synthesis gas (syngas) to light olefins—i.e., $C_2=C_4$ —olefins containing two to four carbon atoms, so called Fischer-Tropsch to olefins (FTTO) (1–5). FTS has been under extensive study for more than 50 years, and a variety of metal catalysts, including iron, cobalt,

and ruthenium, have been tested (6). However, application of this technology is still limited by low olefin selectivity and high methane selectivity, as well as severe carbon deposition. These drawbacks arise from the FTS reaction mechanism, which is generally accepted to proceed via surface polymerization of CH_x ($x = 1, 2$, or 3)—i.e., addition of CH_x monomer units to the adsorbed alkyl species on open metal surfaces. Thus, a wide distribution of hydrocarbons with different chain lengths is produced. It can be described by the Anderson-Schulz-Flory (ASF) model (7, 8), which predicts that selectivity of C_2 to C_4 hydrocarbons (C_2-C_4), including $C_2=C_4$ olefins and $C_2^o-C_4^o$ paraffins, does not exceed 58% (6).

Thus, the key challenge of selective formation of light olefins from syngas lies in precise control of C-C coupling while suppressing overhydrogenation and methane formation. We present here a process named OX-ZEO (Oxide-Zeolite),

which separates CO activation and C-C coupling onto two different types of active sites with complementary properties. CO and H_2 are activated over a partially reduced oxide ($ZnCrO_x$) surface, whereas C-C coupling is controlled within the confined environment of zeolite pores with acidic sites. As a result, a C_2-C_4 selectivity up to 94% (including 80% $C_2=C_4$ and 14% $C_2^o-C_4^o$) of all hydrocarbons (carbon atom-based) was achieved with only 2% methane at a CO conversion of 17%. This C_2-C_4 selectivity is far beyond the maximum predicted by the ASF model in FTS.

The composite catalyst contained an oxide ($ZnCrO_x$) that exhibits a typical spinel structure (fig. S1) and a mesoporous SAPO zeolite (MSAPO) exhibiting CHA structure with a hierarchical pore texture (figs. S1 and S2 and table S1) (9). Figure 1A shows that $C_2=C_4$ selectivity reached 74% at CO conversion of 16% over a catalyst with a mass ratio of $ZnCrO_x/MSAPO = 1.4$, under reaction conditions of $H_2/CO = 1.5$, pressure 2.5 MPa, and space velocity 4800 ml/h-g_{cat}. A higher H_2/CO ratio benefits CO conversion, which rose to 30% at $H_2/CO = 3$, for instance, whereas a higher space velocity favored the selective formation of olefins. Selectivities varied in the range from 67 to 80% for $C_2=C_4$ and 81 to 94% for C_2-C_4 in the studied H_2/CO (0.5 to 3.0) and space velocity ranges (1285 to 7714 ml/h-g_{cat}) and different ratios of $ZnCrO_x/MSAPO$. This $C_2=C_4$ selectivity is higher than the best value reported for FTTO (61%) (4) and exceeds the maximum predicted for C_2-C_4 hydrocarbons according to the ASF distribution in typical FTS, as depicted in Fig. 1B. Both CH_4 and C_5+ selectivities were <5% (Fig. 1A), prominently lower than those in FTTO (in a range of 10 to 40% for CH_4 and >10% for C_5+) (4). In contrast, the C_1 and C_5+ products were difficult to suppress simultaneously in FTTO; causing one to decline caused the other to increase (1, 7). The reactivity of this composite catalyst showed good reproducibility, with CO conversion and selectivity fluctuating within 4 and 5%, respectively, among 11 tests (fig. S3). Furthermore, it delivered rather

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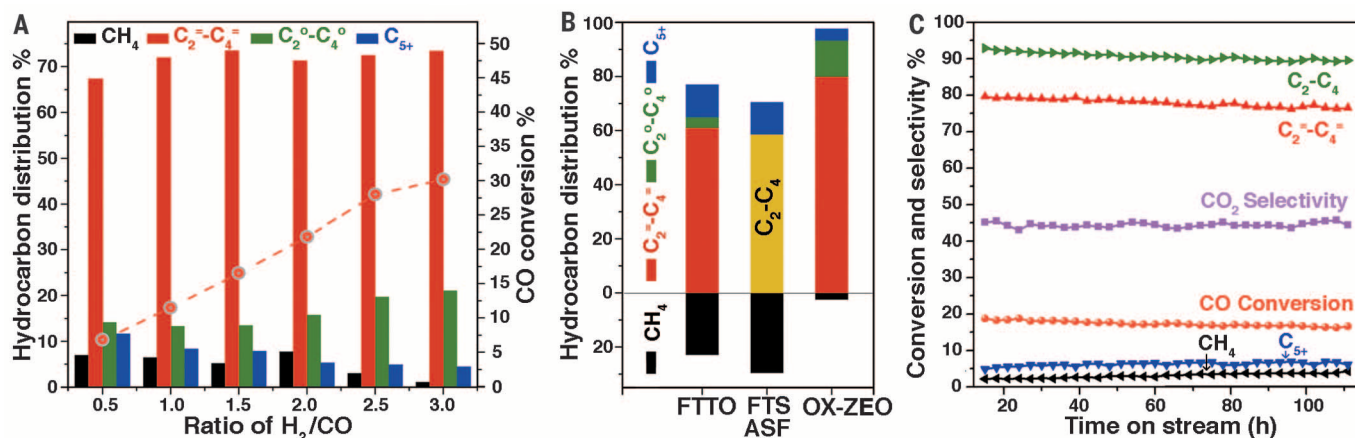


Fig. 1. Catalytic process of OX-ZEO. (A) CO conversion and product distribution at different H_2/CO ratios in syngas over a catalyst with a mass ratio of $ZnCrO_x/MSAPO = 1.4$ at a space velocity of 4800 ml/h-g_{cat}. (B) Hydrocarbon distribution in OX-ZEO in comparison to that reported for FTTO (4) and that in FTS predicted by the ASF model at a chain growth probability of 0.46, with the yellow bar representing selectivity of C_2-C_4 hydrocarbons. (C) A stability test of a composite catalyst with $ZnCrO_x/MSAPO$ ratio = 0.9 at 6828 ml/h-g_{cat} and H_2/CO of 2.5.

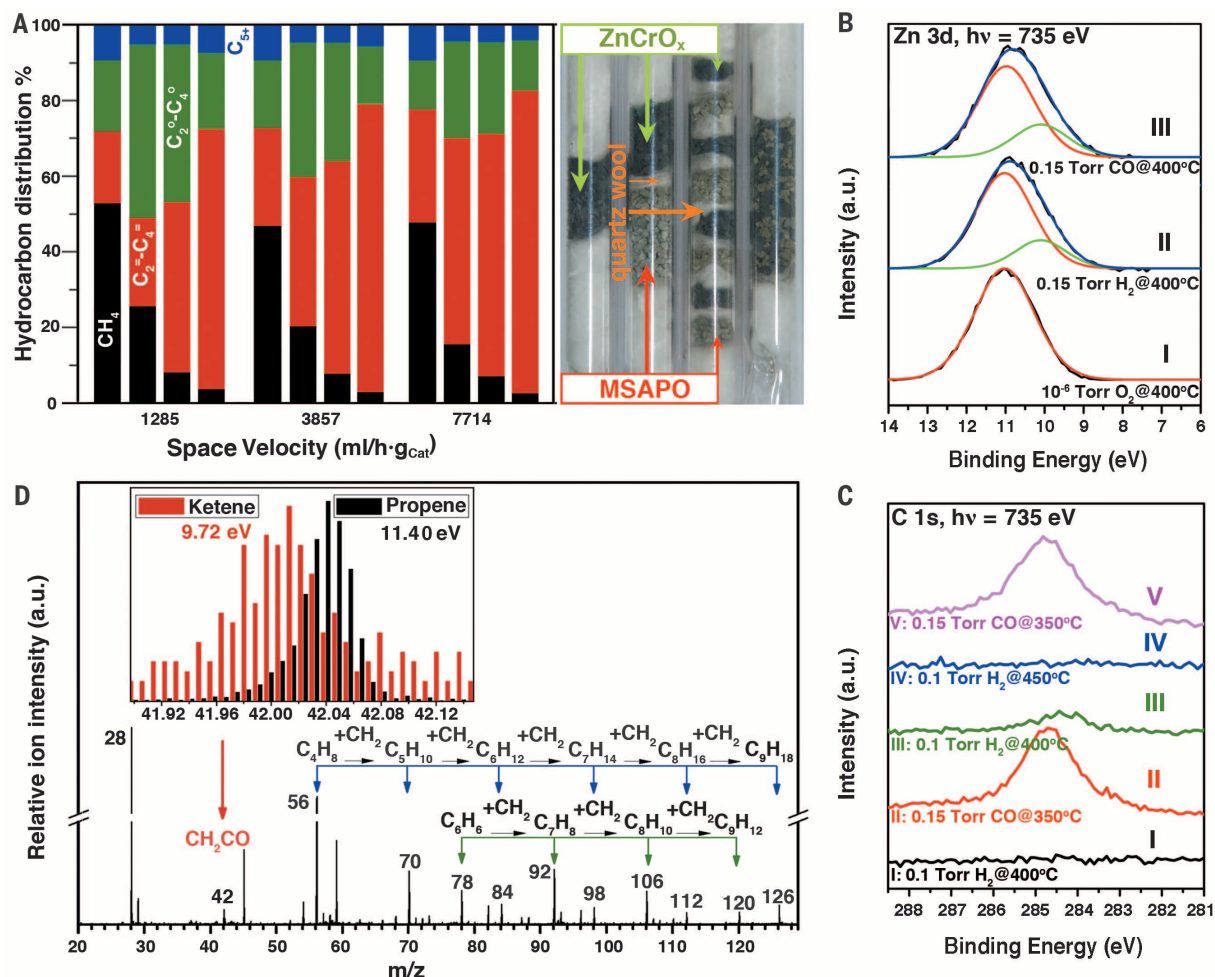


Fig. 2. Bifunctionality of the composite ZnCrO_x/MSAPO catalyst and investigation of the reaction intermediate. (A) Reaction results over the catalysts with the two functionalities packed in different modes under the same conditions. The picture on the right side displays the catalyst beds after reaction, representing Mode 1 to 4 from left to right: Mode 1, the catalyst contains ZnCrO_x only; Mode 2, MSAPO packed below ZnCrO_x, separated by an inert layer of quartz wool; Mode 3, MSAPO and ZnCrO_x packed in an alternating

sequence and separated by quartz wool; Mode 4, MSAPO and ZnCrO_x well mixed. (B) In situ NAP-XPS Zn3d spectra of ZnCrO_x under different conditions. (C) In situ NAP-XPS C1s spectra of ZnCrO_x exposed to H₂, CO, and H₂ sequentially, and then again to CO atmosphere under different conditions. (D) In situ study of syngas conversion over ZnCrO_x by SVUV-PIMS at hν = 9.72 eV. The insets display the signals of m/z = 42.01 (ketene) and m/z = 42.05 (propene) detected at hν = 9.72 and 11.40 eV, respectively.

stable performance (Fig. 1C). The total C₂–C₄ selectivity remained >90%, and that of C₂[–]–C₄[–] ~78% during a 110-hour test at 400°C, 2.5 MPa, and a space velocity of 6828 ml/h·g_{cat}.

Although composite catalysts containing metal oxides such as Cr₂O₃–ZnO and CuO–ZnO, and zeolites such as ZSM-5, Y, and β have been attempted previously for syngas conversion (10–15), the products were mainly dimethylether (10), liquefied petroleum gas (C₃^o–C₄^o) (11–13), or gasoline (14, 15). We attribute the efficiency of the OX-ZEO process to the bifunctionality of the catalyst, with two types of active sites exhibiting complementary and compatible properties. In the absence of MSAPO (corresponding to Mode 1 of Fig. 2A), syngas was converted mainly to CH₄ (selectivity 53%), and selectivity to C₂–C₄ hydrocarbons was only 38% over ZnCrO_x. Upon combination with MSAPO, which was packed below the oxide and separated by a layer of inert quartz wool (Mode 2), the products shifted and C₂–C₄ selectivity increased to

69% containing 23% C₂[–]–C₄[–], whereas CH₄ selectivity dropped to 26%.

This result shows that reaction intermediates generated over the oxides must have transported in the gas phase toward the active sites of MSAPO, where they were converted to C₂–C₄ hydrocarbons, rather than being hydrogenated to CH₄ over the oxide or in gas phase. Thus, any measures that could facilitate the transport of intermediates in gas phase should benefit selective formation of olefins, for example, increasing space velocities, as demonstrated in Fig. 2A. In addition, Mode 3, where the oxides and MSAPO were packed in an alternating sequence and thus the transport distance was reduced, also leads to enhanced C₂[–]–C₄[–] selectivity (65% at 7714 ml/h·g_{cat}). We observed an even higher selectivity of C₂[–]–C₄[–] (80%), with overall C₂–C₄ selectivity reaching 94% and only 2% CH₄ over the well-mixed composite catalyst (Mode 4). Furthermore, CO conversion could be tuned by varying the relative mass ratio of oxide/

zeolite (fig. S4) (9), which lends further support to the above hypothesis that the composite catalyst is bifunctional and that the reaction involves intermediate transport in gas phase.

Activation of the C–O bond has been reported previously for surfaces promoted with oxygen vacancies—for example, Fe₃O₄ (16), CoO_x/TiO₂ (17), Cu/ZnO (0001) (18), and ZnGa₂O₄ (19). However, little is known about subsequent formation of C–C and C–H bonds in the presence of H₂. Density functional theory (DFT) calculations using the ZnCr₂O₄ spinel (111) surface as a model reveal that it is reducible, resulting in a surface with a number of oxygen vacancies (figs. S5 and S6) (9). In situ near-ambient pressure x-ray photoelectron spectroscopy (NAP-XPS) of ZnCrO_x surface (Fig. 2B) revealed a new signal with a lower Zn3d binding energy upon exposure to H₂ and CO, compared with the initial oxidized surface. CO can be activated on such a surface, leading to formation of CO₂, detected by time-of-flight mass spectrometry

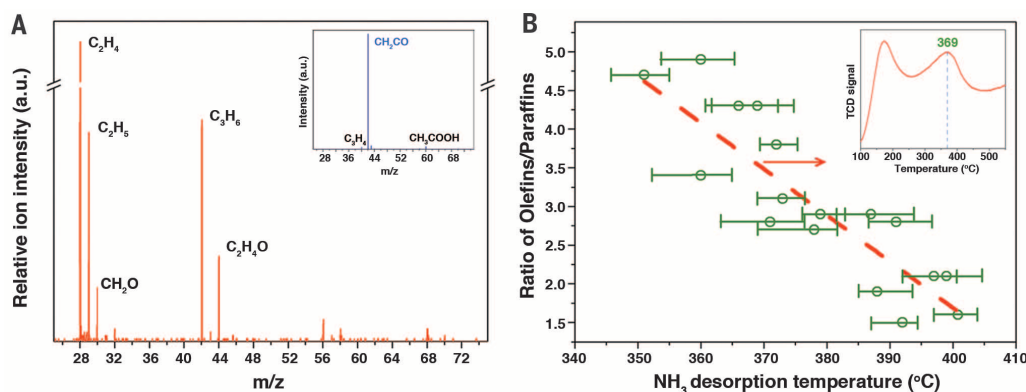


Fig. 3. The role of MSAPO in OX-ZEO.

(A) Catalytic conversion of ketene to olefins over MSAPO at 400°C, with the effluent monitored by SVUV-PIMS. The inset shows the results of the blank experiment with ketene fed to the same reactor containing no catalyst. Spectra are recorded at 10.70 eV. (B) Ratio of olefin/paraffin as a function of the NH_3 desorption peak temperature from the medium-strength acidic sites of MSAPO. The dashed line is only for the purpose of guiding the eyes, and the inset displays a typical NH_3 -TPD profile.

(TOF-MS) (fig. S7) (9) and surface $^*\text{C}$ species in absence of hydrogen, which is evidenced by a strong C1s signal emerging at 284.7 eV, even at 350°C (curve II in Fig. 2C). This C1s signal remained even after heat treatment at 300°C under ultrahigh vacuum (fig. S8) (9), but it remarkably attenuated in H_2 (curve III) and finally vanished at 450°C (curve IV). Analysis of the effluent by TOF-MS shows that $^*\text{C}$ species were removed via hydrogenation forming hydrocarbons (fig. S7, b to d) (9). Thus, the precursor of hydrocarbons, CH_x species formed over ZnCrO_x , are likely the intermediates in the OX-ZEO process.

To detect the intermediate in gas phase, we turned to the highly sensitive synchrotron-based vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) (9), which has been widely used in the field of combustion to detect active intermediates and radicals with low concentrations in gas phase (20, 21). It provides a tunable photon energy, soft ionization, and superior signal-to-noise ratio. The schemes of SVUV-PIMS and the reactor are shown in fig. S9. When syngas was fed into the reactor in the absence of catalysts (a blank experiment), no other signal was observed. In the presence of ZnCrO_x , a signal of mass/charge ratio (m/z) = 42.01 appeared in the effluents at a photon energy $h\nu$ = 9.72 eV (Fig. 2D) in addition to stable hydrocarbon products. Further varying the photon energy (fig. S10) (9) confirmed that this signal was unambiguously attributed to ketene CH_2CO , considering the m/z ratio and the ionization energy (22–24). In comparison, propene (m/z = 42.05) was only distinguishable at $h\nu$ > 9.73 eV (fig. S10e) (9), and it became prominent at $h\nu$ = 11.40 eV (the black signal in the inset of Fig. 2D). Additional signals in Fig. 2D were assigned to stable products of C_4H_8 and C_6H_6 , and their derivatives from stepwise addition of CH_2 monomers up to m/z = 126.

In addition to ketene and the stable products, signals of methanol (m/z = 32) and its dissociated product methoxyl group (m/z = 31) were also detected at a high $h\nu$ (11.40 eV) in the presence of ZnCrO_x catalyst. However, the reaction did not appear to go via methanol in this OX-ZEO process because the yield of C_2^- – C_4^- dramatically dropped to 3% within 22 hours when feeding methanol (50 mbar in He) directly to the composite catalyst (fig. S11). Figure S12 shows that

MSAPO alone as a methanol-to-olefin (MTO) catalyst also deactivated quickly within 30 hours, even at a methanol partial pressure as low as 5 mbar, and 0.18 mol C/g_{cat} was converted before deactivation. By contrast, the composite catalyst delivered rather stable performance in OX-ZEO. No obvious degradation of CO conversion was observed for operation as long as 650 hours (fig. S13), and 9.6 mol C/g_{cat} was converted within this period of time. The total coke deposit was only 11 weight % (wt %) (fig. S14) (9). In contrast, fast deactivation remains a major issue for MTO in that coke deposits can mount up about 10 wt % within 15 min (25). Furthermore, very little water is produced in OX-ZEO, in contrast to the MTO process, where two moles of H_2O are produced for each mole of ethylene (assumed as the only product) from methanol (9). In addition, O from CO is removed mainly by producing CO_2 with a selectivity of 40 to 45% in OX-ZEO, whereas little CO_2 forms in MTO. The above results show that reaction via methanol may not be the dominating pathway in OX-ZEO, although it cannot be excluded completely, whereas ketene likely plays an important role.

Such CH_2 species are very active and readily react with CO in the presence of CO, forming a relatively less reactive ketene (26, 27), which is detectable by SVUV-PIMS. Thus, CH_2 species are likely the primary intermediates. By forming ketene, the reaction pathway of surface polymerization of CH_x is blocked, hence circumventing the ASF limits of FTS. Subsequently, in the presence of confined acidic environment of zeolite pores, CH_2CO can be converted to olefins (Fig. 2A). This was validated by feeding ketene directly as the reactant to the MSAPO catalyst, with the effluents monitored by the online SVUV-PIMS. Ketene was synthesized via pyrolysis of acetic anhydride following the reported procedure (fig. S15) (24, 28). Comparison of the results in the blank reactor and in the presence of the catalyst (Fig. 3A and fig. S15) demonstrates the capability of MSAPO catalyzing conversion of ketene to light olefins.

Figure 3B and fig. S16 (9) demonstrate that the product selectivity can be modulated by the medium strength acidity, characterized by temperature-programmed-desorption (TPD) of NH_3 , with the peak maximum located in the range from 350° to 410°C. For example, a commercially available

SAPO-34 exhibiting NH_3 desorption at ~394°C had a C_2^- – C_4^- selectivity of 43% and an olefin/paraffin ratio (C_2^- – $\text{C}_4^-/\text{C}_5^-$ – C_6^-) of 0.9. The olefin/paraffin ratio increased with the decreasing NH_3 desorption temperature and reached 4.7 for a MSAPO sample with NH_3 desorption peak at 350°C (Fig. 3C). In contrast, a composite catalyst with almost no medium strength acidic sites had a similar product distribution, as the catalyst contained no zeolite (fig. S17 and table S3) (9). Thus, it is reasonable to assume that there may exist an optimum acidity with NH_3 desorption temperature <350°C for olefin formation. However, it is still a challenge to synthesize zeolites with a weaker acidity but without reducing the number of active sites (necessary to achieve a reasonable conversion). In addition, the preliminary results in fig. S18 (9) show that the distribution of different-sized hydrocarbons might be tunable through the shape selectivity of zeolites, because a larger pore generally yields higher hydrocarbons. However, further elucidation of the relation between the structure (pore size/acidity) and the activity/selectivity will require much more sophisticated experiments because the acidity frequently varies simultaneously with the pore structure and crystallinity.

The capability of the partially reduced oxide in activating CO but incapability of catalyzing surface polymerization of CH_x makes it possible to manipulate C–C coupling with confined acidic zeolite pores. Thus, the CO conversion and selectivity can be tuned at the same time—i.e., CO conversion is manipulated via the surface structure of the oxides and the ratio of oxides/zeolite, whereas the olefin selectivity is controlled by the properties of zeolites, particularly the pore structure and acidity. These findings open up a new avenue for development of syngas-to-olefin technologies, which may allow utilization of coal- and biomass-derived syngas with a low H_2/CO ratio.

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SUPPLEMENTARY MATERIALS

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QUANTUM COMPUTING

Realization of a scalable Shor algorithm

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Certain algorithms for quantum computers are able to outperform their classical counterparts. In 1994, Peter Shor came up with a quantum algorithm that calculates the prime factors of a large number vastly more efficiently than a classical computer. For general scalability of such algorithms, hardware, quantum error correction, and the algorithmic realization itself need to be extensible. Here we present the realization of a scalable Shor algorithm, as proposed by Kitaev. We factor the number 15 by effectively employing and controlling seven qubits and four “cache qubits” and by implementing generalized arithmetic operations, known as modular multipliers. This algorithm has been realized scalably within an ion-trap quantum computer and returns the correct factors with a confidence level exceeding 99%.

Shor’s algorithm for factoring integers (*1*) is one example in which a quantum computer (QC) outperforms the most efficient known classical algorithms. Experimentally, its implementation is highly demanding (*2–7*) because it requires both a sufficiently large quantum register and high-fidelity control. Such challenging requirements raise the question of whether optimizations and experimental shortcuts are possible. Optimizations, especially system-specific or architectural optimizations, are certainly possible, but for a demonstration of Shor’s algorithm in a scalable manner, special care must be taken to not oversimplify the implementation—for instance, by employing knowledge about the solution before the actual experimental application (*8*).

How does Shor’s algorithm work? First, we consider a classical factoring recipe, assuming

that the number we want to factor is $N = 15$. We pick a random number $a \in [2, N - 1]$ (the base)—say, $a = 7$. We evaluate whether the greatest common divisor $\text{gcd}(a, N) = 1$; if not, a factor is already determined. This is the case for $a = \{3, 5, 6, 9, 10, 12\}$. Next, we calculate the modular exponentiation $a^x \bmod N$ for $x = 0, 1, 2, \dots$ and find its period r : the first value of $x > 0$ such that $a^x \bmod N = 1$. Given r , finding the factors of N requires calculating the greatest common divisors of $a^{r/2} \pm 1$ and N , which is efficiently possible with a classical approach—for instance, using Euclid’s algorithm. For our example ($N = 15$, $a = 7$), the modular exponentiation yields 1, 7, 4, 13, 1, ..., which has a period of 4. The greatest common divisors of $a^{r/2} \pm 1 = 7^{4/2} \pm 1 = \{48, 50\}$ and $N = 15$ are $\{3, 5\}$, the nontrivial factors of N . In this example, the cases $a = \{4, 11, 14\}$ have period $r = 2$ and require a single multiplication step ($a^2 \bmod N = 1$), which is considered an “easy” case (*8*). Note that the periodicity for a chosen a cannot be predicted.

How can this recipe be implemented in a QC? A QC also has to calculate $a^x \bmod N$ in a computational register for $x = 0, 1, 2, \dots$ and then extract r . Using the quantum Fourier transform (QFT) applied to the period register, the period of

$a^x \bmod N$ can be extracted from a number of measurements not increasing with the size of the number to be factored.

What are the requirements and challenges of implementing Shor’s algorithm? We first focus on the period register and subsequently address modular exponentiation in the computational register. Factoring N , an $n = \lceil \log_2(N) \rceil$ -bit number (with the quantity in brackets rounded up to next integer number), requires a minimum of n qubits in the computational register (to store the results of $a^x \bmod N$) and generally about $2n$ qubits in the period register (*9, 10*). Thus, even a seemingly simple example, such as factoring 15 (an $n = 4$ -bit number), requires $3n = 12$ qubits. These qubits then have to be manipulated with high-fidelity gate operations. Given the current state-of-the-art control over quantum systems (*11*), such an approach would probably yield an unsatisfactory performance. However, a full quantum implementation of this part of the algorithm is not necessary. As noted by Kitaev (*12*), if only the classical information of the QFT (such as the period r) is of interest, $2n$ qubits subject to a QFT can be replaced by a single qubit. Still, this approach requires qubit recycling (specifically, in-sequence single-qubit readout and state reinitialization) paired with feed-forward behavior to compensate for the reduced system size.

In the following, Kitaev’s QFT will be referred to as KQFT^(M). It replaces a QFT acting on M qubits with a semiclassical QFT acting repeatedly on a single qubit. Similar applications of Kitaev’s approach to a semiclassical QFT in quantum algorithms have been investigated (*13–15*). For the implementation of Shor’s algorithm, Kitaev’s approach provides a reduction from the previous n computational qubits and $2n$ QFT qubits (in total, $3n$ qubits) to only n computational qubits and 1 KQFT⁽²ⁿ⁾ qubit (in total, $n + 1$ qubits).

The second key ingredient of Shor’s algorithm—and a notably more challenging aspect than the QFT—is modular exponentiation, which admits the following general simplifications.

1) Considering Kitaev’s approach (Fig. 1), the input state $|1\rangle$ (in decimal representation) is subject to a conditional multiplication based on the most significant bit k of the period register. At most, there will be two results after this first step.

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Supplementary Materials for

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Materials and Methods

Catalyst preparation

ZnCrO_x oxide was prepared by co-precipitation method. Briefly, 14.56 g Zn(NO₃)₂ · 6H₂O, 5.6 g Cr(NO₃)₃ · 9H₂O, 5.25 g Al(NO₃)₃ · 9H₂O were dissolved in 100 ml distilled water and (NH₄)₂CO₃ aqueous solution was used as the precipitant. Precipitation was allowed at 70 °C, followed by aging for 3 h at the same temperature. After filtering and washing by distilled water, the resulting product was dried overnight at 110 °C and then calcined at 500 °C for 1 h.

Zeolite MSAPO was synthesized hydrothermally. Typically, 30% silica sol, 82% AlOOH, 85% phosphoric acid and triethylamine (TEA) were well dispersed in distilled water with a ratio of SiO₂:Al₂O₃:H₃PO₄:TEA:H₂O = 0.48:1:1.6:3:50. Then the mixture was placed in a Teflon-lined autoclave, and kept for 12–28 h in a temperature range of 180–200 °C. The resulting solid product was washed by centrifuging till pH of the supernate was 7.0–7.5. After drying for over 12 h at 110 °C, it was calcined for 3 h at 550 °C.

Commercial SAPO-34, ZSM-5 (Si/Al ratio = 80) and Beta zeolite (Si/Al ratio = 40) were purchased from Nankai University Catalyst Company, China.

Catalytic reaction tests

Catalytic reactions were performed in a continuous flow, fixed-bed stainless steel reactor furnished with a quartz lining. Typically 280–300 mg composite catalyst (20–40 mesh) with ZnCrO_x/MSAPO = 1/1 (mass ratio) was used unless otherwise stated. For a test with the oxide ZnCrO_x without MSAPO as Mode 1 of Figure 2A, 140 mg catalyst was used. The catalyst was subjected to in situ reduction in H₂ for 2 h at 310 °C prior to reaction. Syngas with CO and H₂ contains 5% Ar as the internal standard for online gas chromatography (GC) analysis. Reaction was carried out under conditions of H₂/CO = 2.5, 2.5 MPa, 400 °C and 5143 ml/h g_{cat} unless otherwise stated. Data were collected after at least 4 hours on stream.

A test with methanol as the feeding was also performed in a continuous flow, fixed-bed reactor at 400 °C and atmospheric pressure, over the composite catalyst and also just MSAPO. After reduction in H₂ at 310 °C for 2 h, methanol was introduced with He as the carrier gas. The weight hourly space velocity of methanol was set at 0.5 h⁻¹, equivalent to the yield of olefins of the OX-ZEO process assuming 100% methanol conversion.

Products were analyzed by an online GC (Agilent 7890B), which was equipped with a thermal conductivity detector (TCD) and a flame ionization detector (FID). Hayesep Q and 5Å molecular sieves packed columns were connected to TCD while HP-FFAP and HP-AL/S capillary columns were connected to FID. Oxygen-containing compounds and hydrocarbons up to C₁₇ were analyzed by FID, while CO, CO₂, CH₄, C₂H₄, and C₂H₆ were analyzed by TCD. CH₄, C₂H₄ and C₂H₆ were taken as a reference bridge between FID and TCD. CO conversion was calculated on a carbon atom basis, i.e.

$$Con_{CO} = \frac{CO_{inlet} - CO_{outlet}}{CO_{inlet}} \times 100\%$$

Where CO_{inlet} and CO_{outlet} represent moles of CO at the inlet and outlet, respectively. CO₂ selectivity (Sel_{CO_2}) was calculated according to:

$$Sel_{CO_2} = \frac{CO_{2outlet}}{CO_{inlet} - CO_{outlet}}$$

Where $CO_{2outlet}$ denotes moles of CO₂ at the outlet.

The selectivity of individual hydrocarbon C_nH_m ($Sel_{C_nH_m}$) was obtained according to:

$$Sel_{C_nH_m} = \frac{nC_nH_{moutlet}}{\sum_1^n nC_nH_{moutlet}} \times 100\%$$

The selectivity to oxygenates was below 0.5%C and therefore was not reported in the product selectivity. The carbon balance was over 95%.

In situ near ambient pressure X-ray photoelectron spectroscopy (NAP-XPS)

In situ near ambient pressure X-ray photoelectron spectroscopy (NAP-XPS) was carried out at the beamline 9.3.2 at the Advanced Light Source, Berkeley. The catalyst

was pressed into pellets, which were first degassed under ultrahigh vacuum (UHV) around 200 °C and then oxidized in 1×10^{-6} Torr O₂ to remove any carbon contamination at the catalyst surface. High-purity CO and H₂ were allowed into the analysis chamber with the partial pressure of CO up to 0.5 Torr and H₂ up to 0.15 Torr. Zn 3d, Cr 2p, O 1s, C 1s, Al 2s, and Si 2p spectra were in-situ acquired using the photon energy of 735 eV. The samples show a strong charging effect below 250 °C. However, above 300 °C and in the presence of high pressure gases there is not much influence. Al 2s and Si 2p were used as the references to calibrate all binding energy positions.

Synchrotron-based VUV photoionization mass spectrometry (SVUV-PIMS)

Synchrotron-based VUV photoionization mass spectrometry (SVUV-PIMS) study was carried out at the combustion beamline of the National Synchrotron Radiation Laboratory at Hefei, China. As shown in fig. S9, a quartz reactor with a nozzle size of ~0.1 mm was designed, which was connected to the online SVUV-PIMS spectrometer. The detection limit of SVUV-PIMS is 0.1 ppm for CO₂. Atmospheric pressure syngas (H₂/CO = 2.5) was introduced in and the reaction was allowed in the temperature range of 355 – 400 °C.

Ketene synthesis and catalytic conversion of ketene to light olefins

Ketene was synthesized via pyrolysis of acetic anhydride according to the previously reported procedure (24, 28). Control experiments were carried out to check the catalysis of MSAPO in ketene conversion to light olefins by feeding ketene directly to the MSAPO catalyst at 400 °C and the effluents were monitored by SVUV-PIMS.

Catalyst characterization

X-ray diffraction (XRD) was measured on a PANalytical X'pert PPR diffractometer equipped with CuK α radiation source ($\lambda = 1.5418$ Å), operated at 40 mA and 40 kV. XRD patterns were recorded in the range of $2\theta = 5$ –80 °.

(S)TEM experiments were carried out on the FEI F30 transmission electron microscopes with a resolution of 0.17 nm operated at an acceleration voltage of 300 kV, and JEM-ARM200F TEM with a probe corrector working at 200 kV.

Temperature-programmed-desorption of NH_3 (NH_3 -TPD) was performed on an Micromeritics AutoChem 2910 equipped with a thermal conductivity detector (TCD). The catalyst was first pretreated in a flowing Ar at 450 °C for 1.5 h. After cooling down to 100 °C under flowing Ar, the sample was exposed to 15% NH_3/He at 100 °C. Then the sample was swept by He at 100 °C till a stable baseline was obtained. Subsequently, signal was recorded while the temperature was increased from 100 to 800 °C at a heating rate of 10 °C/min.

Temperature-programmed-reduction of CO and the following surface reaction with H_2 was also performed with the same setup of SVUV-PIMS. After in situ H_2 reduction for 2 h at 310 °C followed by degasing to vacuum down to around $(4-6) \times 10^{-5}$ Pa at 25 °C. CO was allowed in at a flow rate of 12.7 ml/min. Subsequently, the catalyst was heated to 400 °C at a rate of 10 °C/min and held for 15 min at this temperature. The effluent was monitored at a photon energy $h\nu = 14.50$ eV, which allowed detection of both CO and CO_2 . Subsequently, the catalyst was cooled down to 25 °C and evacuated again. Then H_2 was introduced and the temperature was increased to 400 °C at the same rate. The effluent was monitored at $h\nu = 13.00$ eV, which allowed detection of hydrocarbons including methane (ionization energy = 12.61 eV).

Nitrogen adsorption-desorption was carried out on a Quantachrome NOVA 4200e. Before analysis, all samples were degassed under vacuum at 300 °C for 6 h. Isotherms were recorded at 77 K.

Thermogravimetric Analysis (TGA) was conducted on Pyris Diamond TG-DTG in a flowing air (100 ml/min) and the heating rate was 10 °C/min from 30 to 900 °C.

Theoretical calculations

Theoretical calculations were carried out based on Density Functional Theory (DFT) framework by using Vienna ab initio simulation packages (VASP) with the projector

augmented wave (PAW) scheme (29). The Perdew-Burke-Ernzerhof (PBE) functional was adopted for the exchange-correlation effect (30). A model of spinel ZnCr_2O_4 (111) surface with two repeating unit layers (12 atomic layers) and the vacuum thickness of about 20 Å were used to study all adsorption and other kinetic processes. The Brillouin zone was sampled at gamma. All atoms except the atomic layers in the bottom were relaxed until the forces are smaller than 0.05 eV/Å.

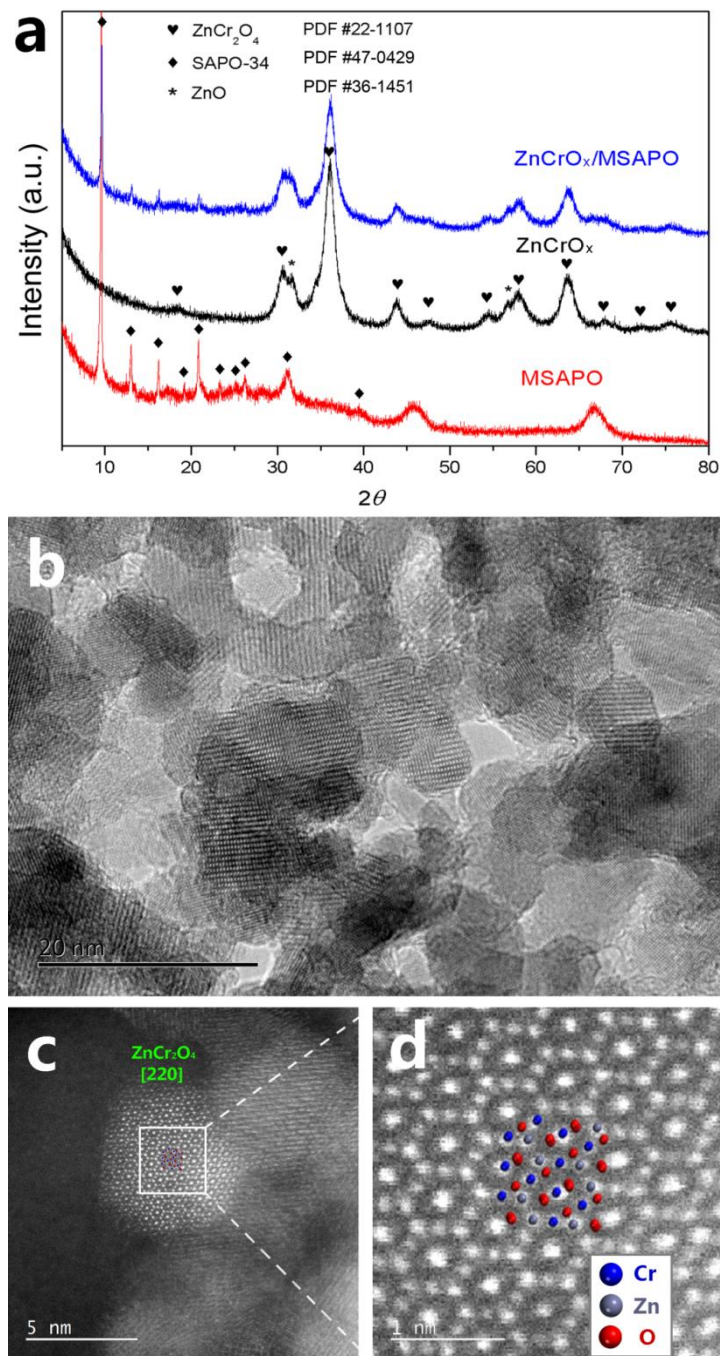


Fig. S1.(a) XRD patterns of ZnCrO_x , MSAPO and the resulting composite catalyst. ZnCrO_x exhibits a typical spinel structure, corresponding to PDF #22-1107. The diffraction peaks of MSAPO matches with the characteristic diffraction of SAPO-34. (b)-(d) High resolution (S)TEM images of ZnCrO_x at different regions of the specimen with (c) and (d) showing the atomic structure of spinel.

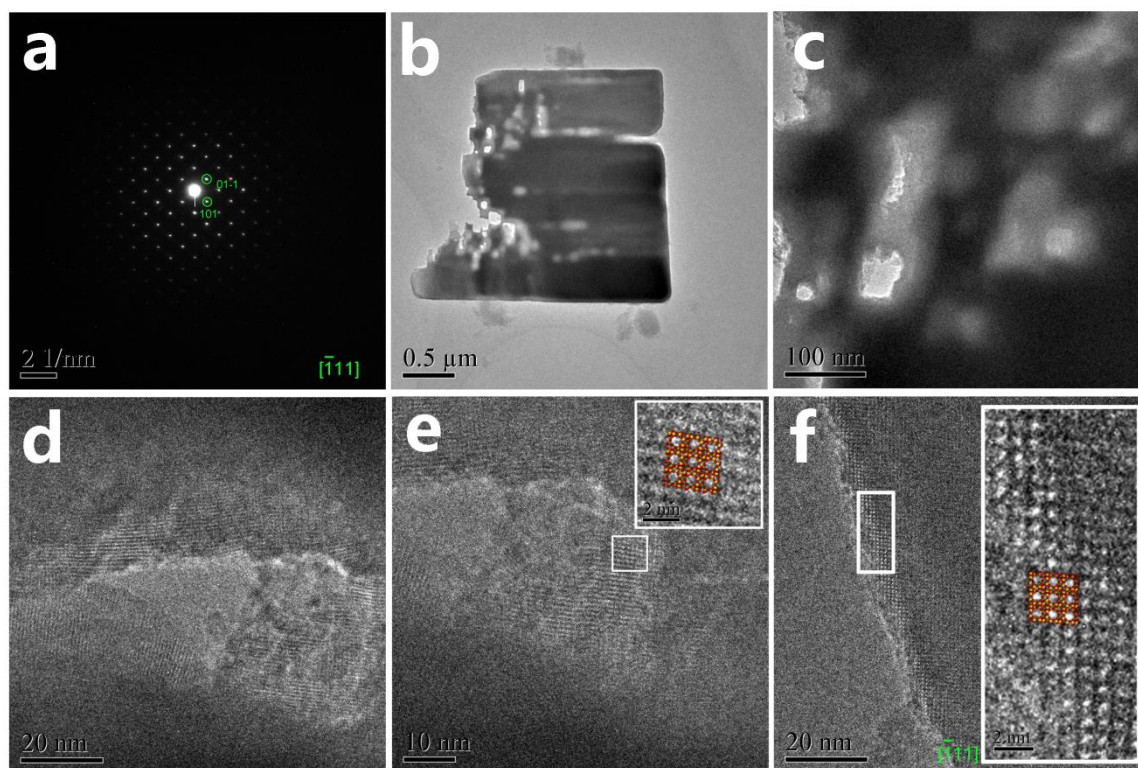


Fig. S2 TEM images of MSAPO. (a) Electron diffraction pattern showing typical SAPO-34 structure observed along the $[-111]$ axis. (b)–(f) Typical morphology of MSAPO with (c)–(e) stepwise enlarged images, which exhibit clearly a hierarchical pore structure with macro-, meso- and micropores. The lattice structure is also clearly observed in (d)–(f) corresponding to SAPO-34. The insets in (e) and (f) show the enlarged image of the regions marked by the white rectangles and the pore structures fit very well with that of characteristic CHA crystal with micropores observed along the $[-111]$ axis (denoted with yellow and red colors).

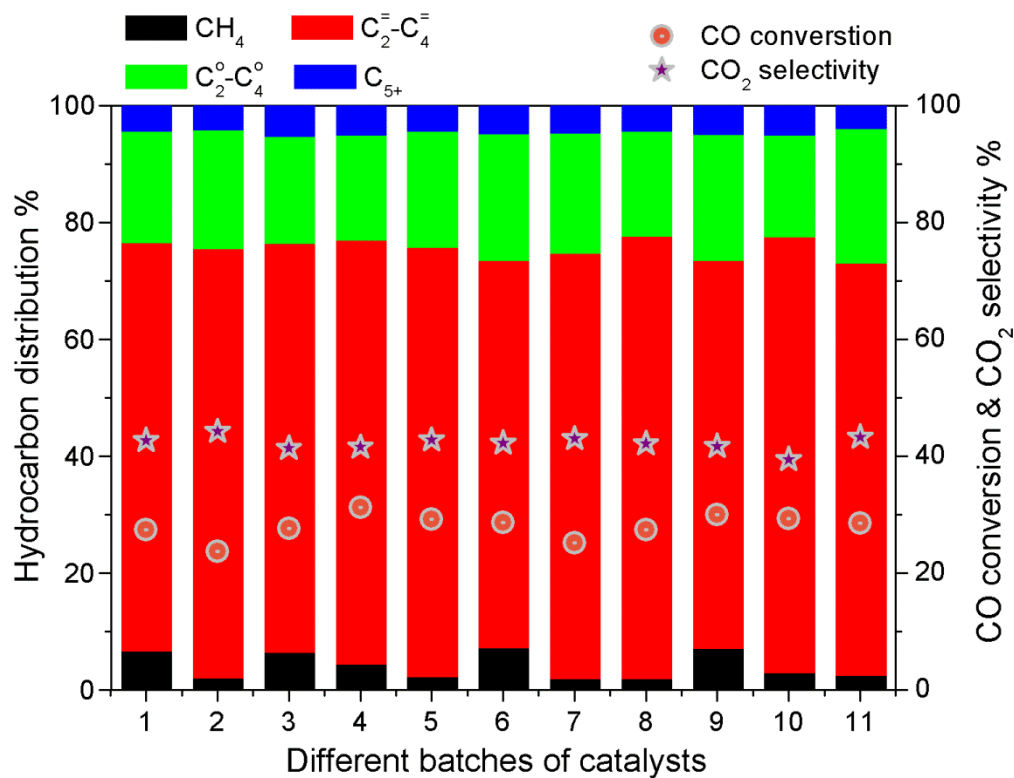


Fig. S3 Reproducibility test of catalysts, which have been prepared and evaluated in different batches under the same reaction conditions. The results demonstrate that the reactivity is well reproducible. Reaction conditions: 400 °C, 2.5 MPa and GHSV 5143 ml/h g_{cat}.

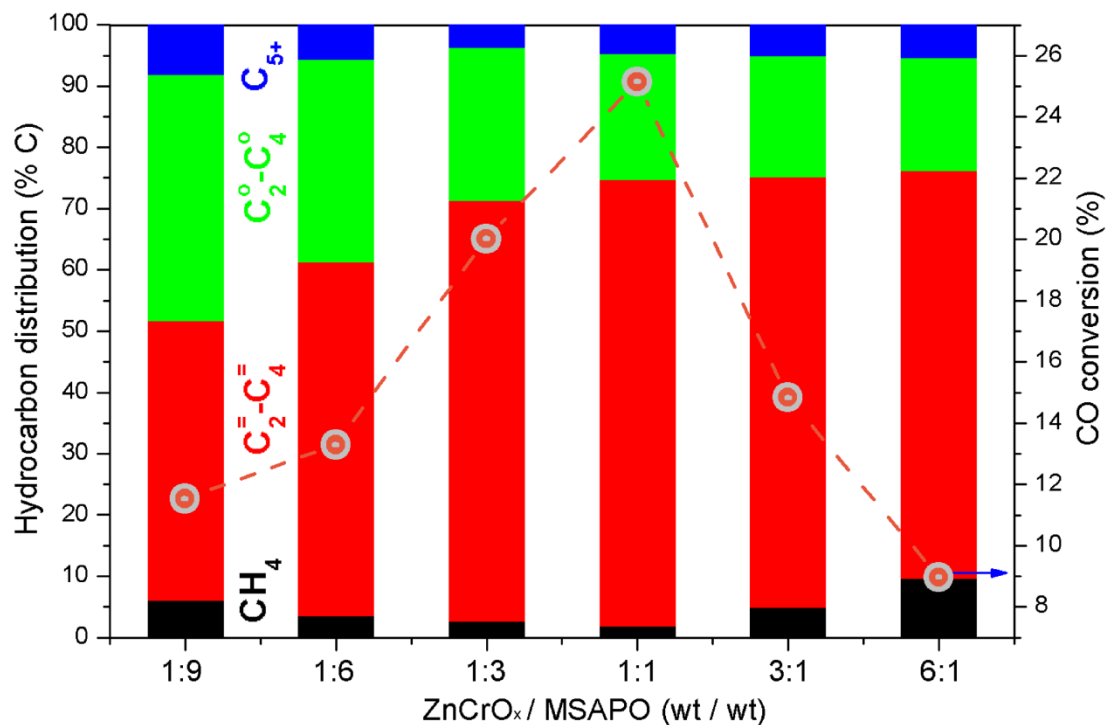


Fig. S4 Reaction results over composite catalysts as a function of the mass ratios of ZnCrO_x/MSAPO in the composite catalysts, which lend further support to the hypothesis that the composite catalyst is bifunctional and that the reaction involves intermediate transport in gas phase. Reaction conditions: 400 °C, 2.5 MPa and GHSV 5143 ml/h g_{cat}.

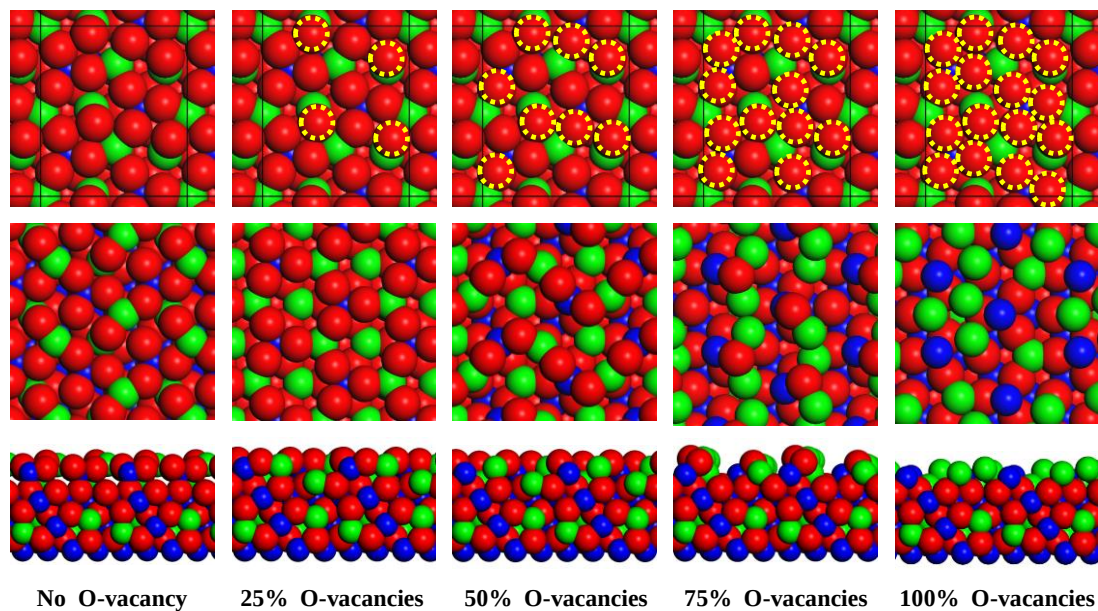


Fig. S5 Optimized models for spinel ZnCr_2O_4 containing different concentration of surface oxygen vacancies of 0%, 25%, 50%, 75% and 100% O at the first atomic layer, respectively. The top row shows the surface before optimization with the removed oxygen atoms marked in yellow dashed circles; the second row and the third row show the surface and side views of the optimized spinel ZnCr_2O_4 surface containing vacancies. Zn: green; Cr: blue; O: red.

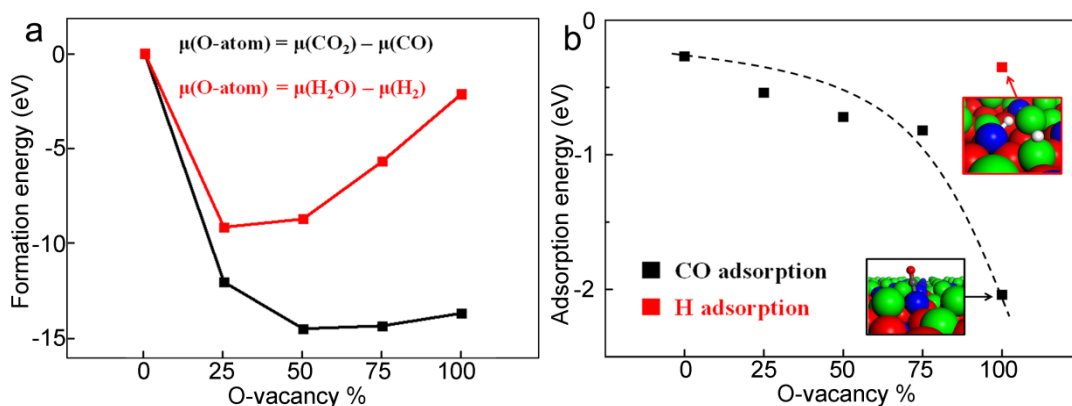


Fig. S6 (a) DFT calculations show that spinel ZnCr₂O₄ (111) surface is reducible and reduction by CO is much more energy favorable than by H₂, which leads to a partially reduced surface containing various concentrations of oxygen vacancies. The chemical potential of oxygen on the surface is assumed to be directly related with that of CO and CO₂ (black) for reduction by CO, and H₂O (red) for reduction by H₂, respectively. (b) Calculated adsorption energies for CO (black squares) and H atoms (red square) on the spinel ZnCr₂O₄ (111) surface with different concentration vacancies. The optimized CO and H₂ adsorption configurations on the surface with 100% O vacancies at the first atomic layer are shown in the insets. Zn: green; Cr: blue; O: red; H: white; C: grey.

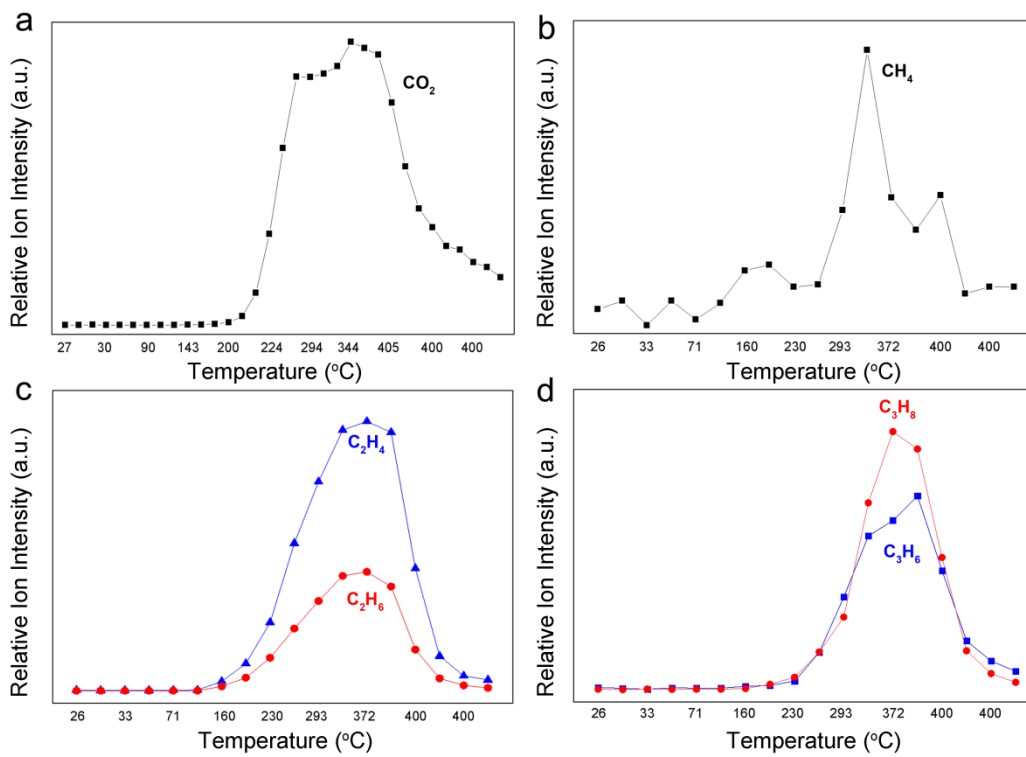


Fig. S7 (a) Temperature-programmed-reduction of ZnCrO_x by CO, with the effluent monitored by SVUV-PIMS. The results show that CO₂ was released with the temperature increasing. Extended exposure to CO at 400 °C leads to dissociation of CO forming CO₂ and surface *C species, evidenced by in situ NAP-XPS. The CO₂ yield declines with time due to increasing surface coverage of *C species. (b)–(d) Subsequent introduction of H₂ leading to formation of hydrocarbons due to hydrogenation of surface *C species. Hydrocarbon signals with *m/z* up to 126 are detected and those containing up to 3 carbon atoms are shown here as examples.

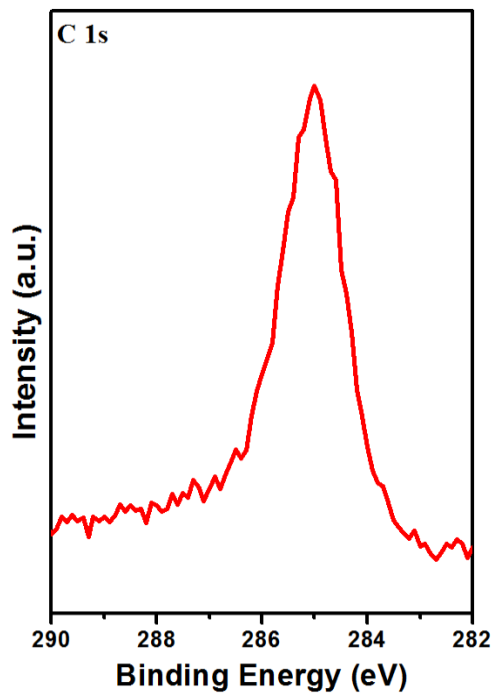


Fig. S8 In-situ NAP-XPS C1s spectrum of ZnCrO_x recorded after evacuation in UHV at 300 °C following exposure to CO, indicating that this species is surface *C species from dissociated CO rather than surface adsorbed CO. The experiments were carried out at the Advanced Light Source, Berkeley.

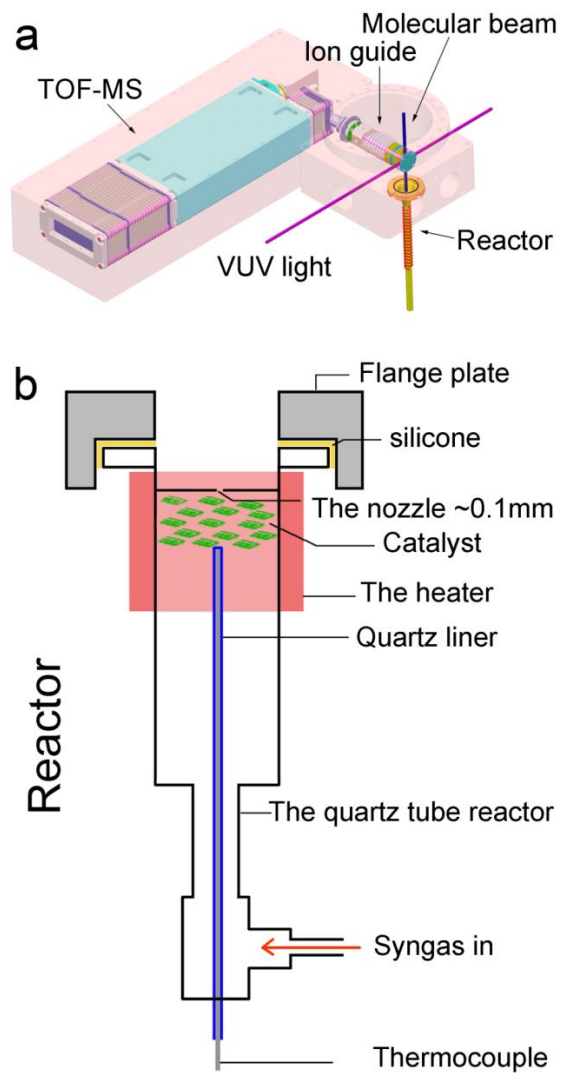


Fig. S9 Schemes for (a) the SVUV-PIMS system and it is connected with a catalytic reactor. (b) the catalytic reactor. The experiments were carried out at the combustion beamline of the National Synchrotron Radiation Laboratory at Hefei, China.

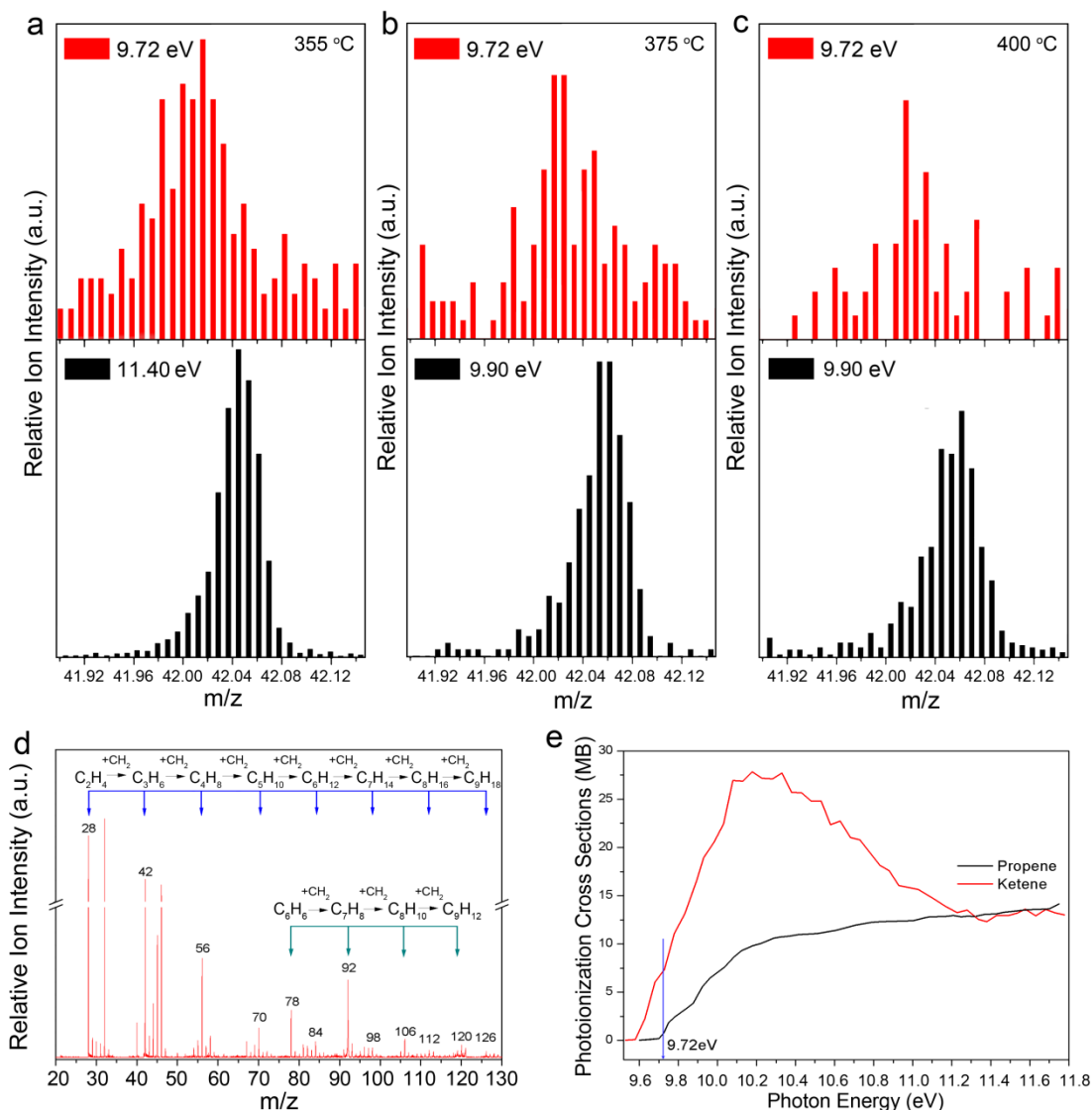


Fig. S10 In situ SVUV-PIMS study of syngas conversion over ZnCrO_x . (a) Signals of $m/z = 42.01$ (ketene) and $m/z = 42.05$ (propene), detected at 355 °C and different photon energies ($h\nu = 9.72$ and 11.40 eV , respectively). (b) Signals of ketene and propene, detected at 375 °C , and $h\nu = 9.72$ and 9.90 eV , respectively. (c) Signals of ketene and propene detected at 400 °C , and $h\nu = 9.72$ and 9.90 eV , respectively. Although the intensity of ketene declines with the increasing temperature likely because of its higher reactivity, it is still discernible at 400 °C . The $m/z = 42.01$ is unambiguously attributed to ketene CH_2CO considering the m/z ratio and the ionization energy. (d) A full spectrum collected at 355 °C and $h\nu = 11.40 \text{ eV}$. Note that the intensity of ions is proportional to the concentration and the photoionization cross section. In addition to ketene and stable

products of hydrocarbons, methanol ($m/z = 32$, ionization energy 10.84 eV) with a small amount of the dissociated product methoxyl group ($m/z = 31$) are also detected at this high $h\nu$. However, over the composite catalyst the reaction does not likely go via methanol or methoxyl group because feeding methanol directly to the composite catalyst lead to dramatically dropping yield of $C_2^+=C_4^+$ down to 3% within 22 h in contrast to the stable performance in OX-ZEO as long as 650 h. MSAPO alone as a MTO catalyst also deactivates within 30 h even at a methanol partial pressure of 5–20 mbar. Furthermore, there is very little water produced in OX-ZEO process in contrast to MTO. (e) Absolute photoionization cross sections for propene (23) and ketene (24).

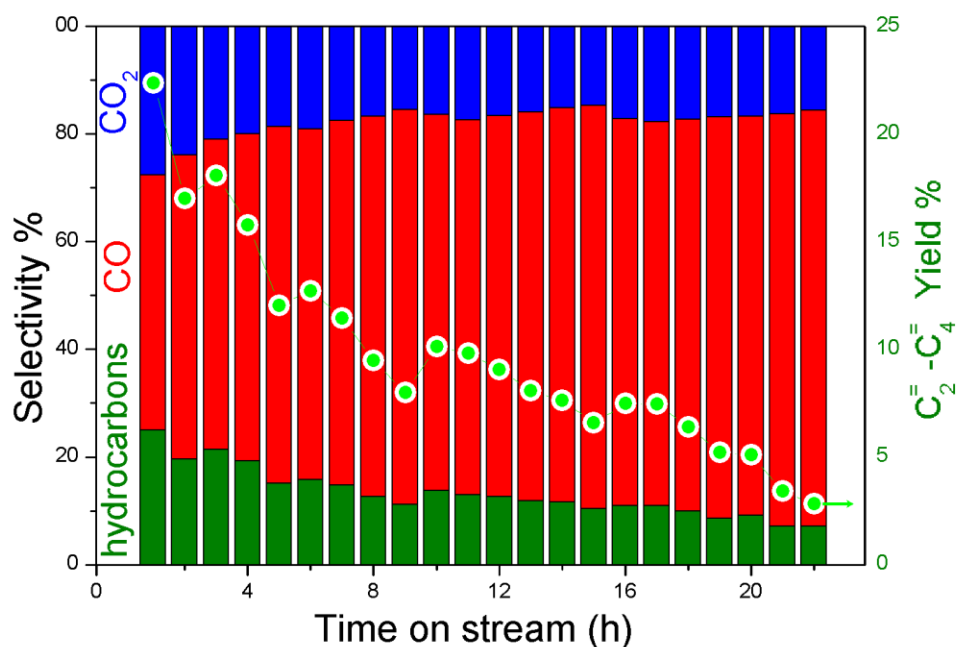


Fig. S11 Reaction results over the composite catalyst by feeding methanol (50 mbar in He) as the reactant at 400 °C, WHSV of 0.5 h⁻¹ and atmospheric pressure. The green filled circles represent the yield of C₂=-C₄= and filled bars represent the selectivities of different products. A high selectivity of CO indicates that methanol mainly decomposes while the rest methanol goes through MTO reaction. However, the C₂=-C₄= yield drops dramatically within 22 h and the selectivity to hydrocarbons also decreases quickly, indicating fast deactivation of the composite catalyst in MTO in contrast to the stable performance in syngas conversion in a 650 h test.

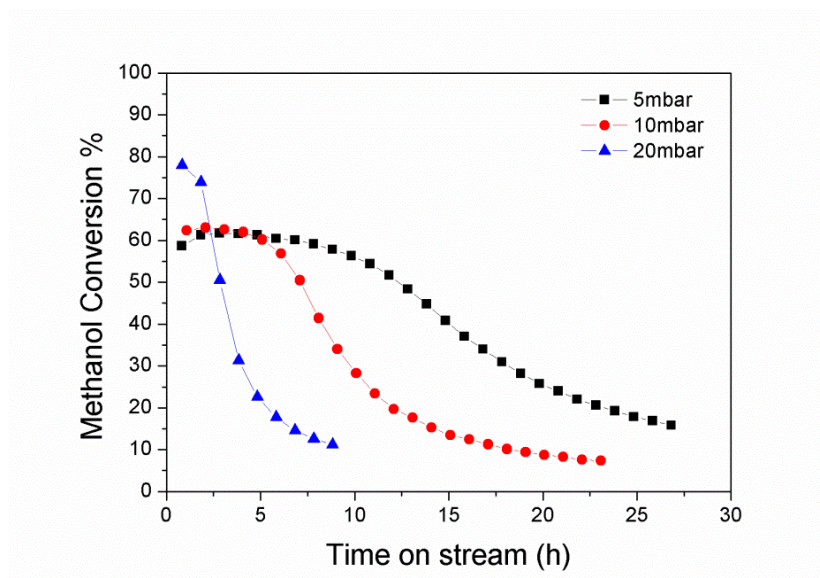


Fig. S12 Reaction results over the MSAPO catalyst by feeding methanol as the reactant at 400 °C at varying methanol partial pressures, 5, 10 and 20 mbar using He as the carrier gas. The amount of methanol feeding was kept at a value to achieve the same olefin yield as in OX-ZEO process assuming 100% conversion of methanol. The results show quick deactivation of the catalyst around 20 h at 10 mbar and within less than 30 h at an even lower partial pressure of 5 mbar converting 0.18 mol C/g_{cat} before deactivation. This is in contrast to the stable performance as long as 650 h in our OX-ZEO process and 9.6 mol C/g_{cat} was converted within this period of time. Furthermore, methanol conversion far lower than 100% indicates that not all olefins are produced from methanol. These results suggest that methanol is not the dominating reaction pathway. There should exist another reaction pathway.

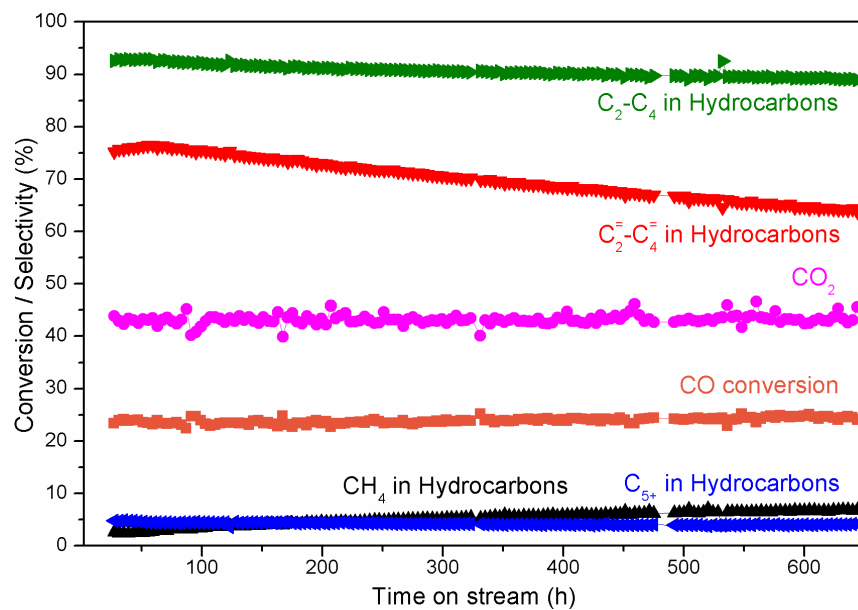


Fig. S13 A stability test for 650 h under condition of $H_2/CO = 2.5$, 400 °C and 5143 ml/h g_{cat} . This catalyst has an oxide/zeolite mass ratio of 1/3.

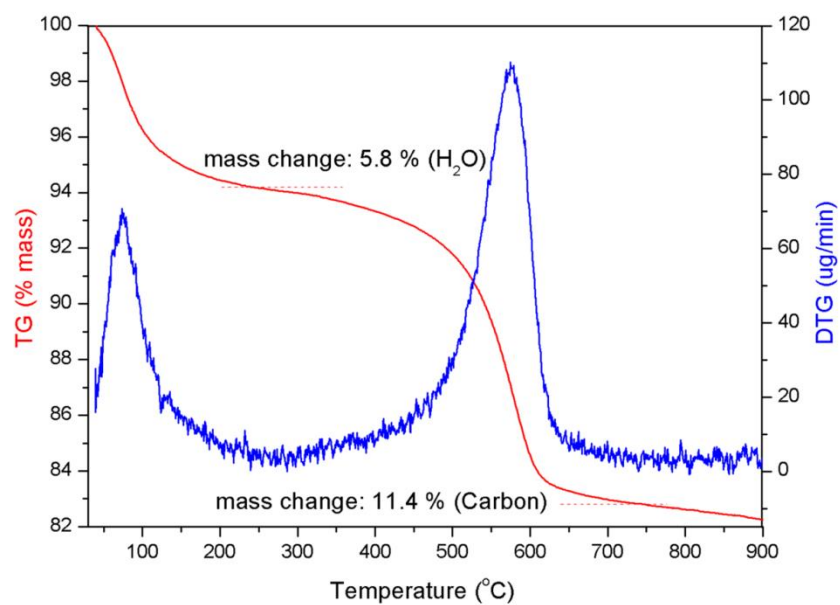


Fig. S14 TG & DTG analysis of the composite catalysts after reaction tests of 650 h in Fig. S13.

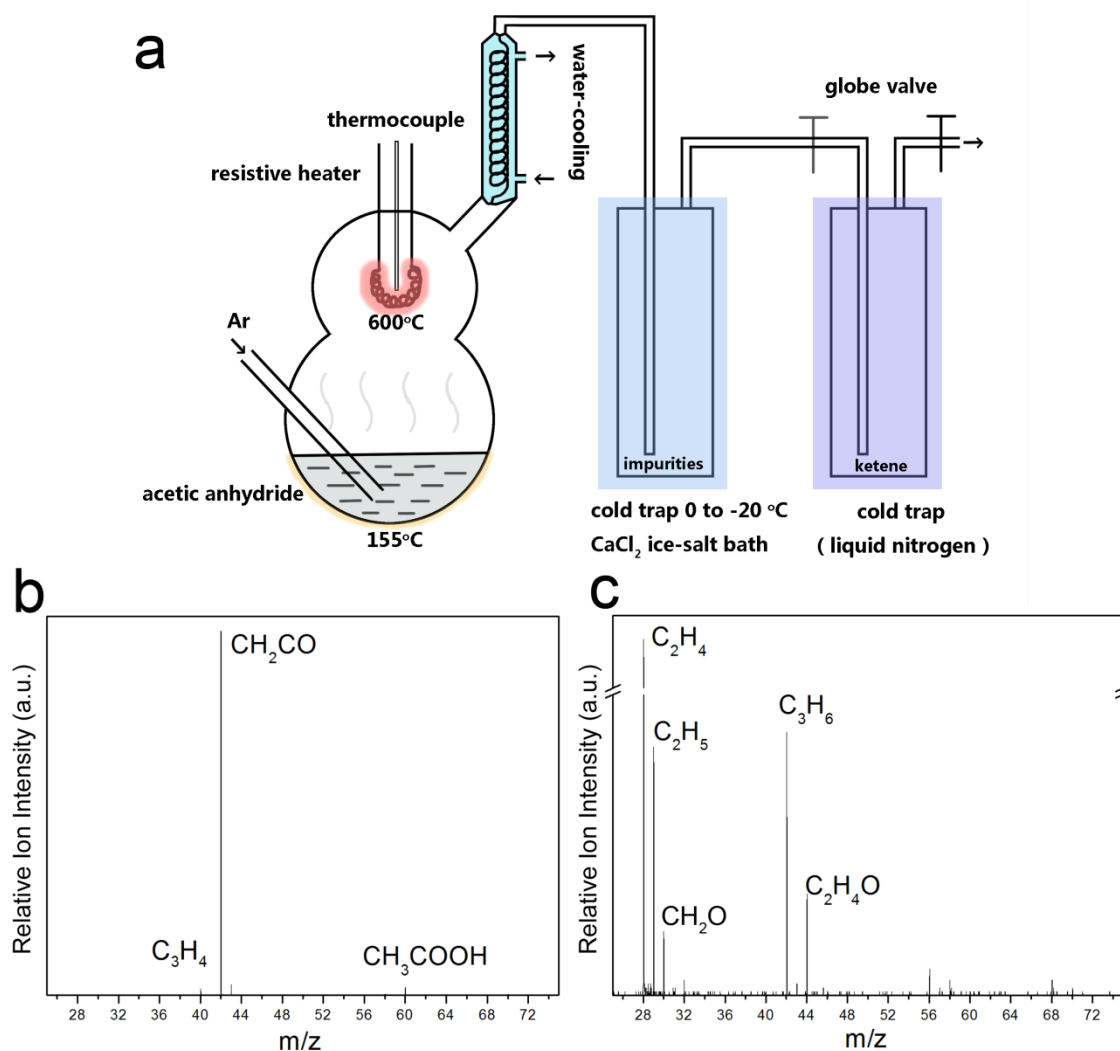


Fig. S15 (a) Experimental setup for synthesis of ketene from acetic anhydride via pyrolysis following the reported procedure (24, 28); (b) Blank experiment with ketene fed to the empty reactor in the absence of the catalyst. The strong peak of $m/z = 42.01$ corresponds to ketene while the other smaller peaks are attributed to impurities in ketene. (c) Results with ketene fed to the MSAPO catalyst. Reaction temperature was 400 °C with the effluents monitored by the online SVUV-PIMS and the spectra were recorded at 10.70 eV.

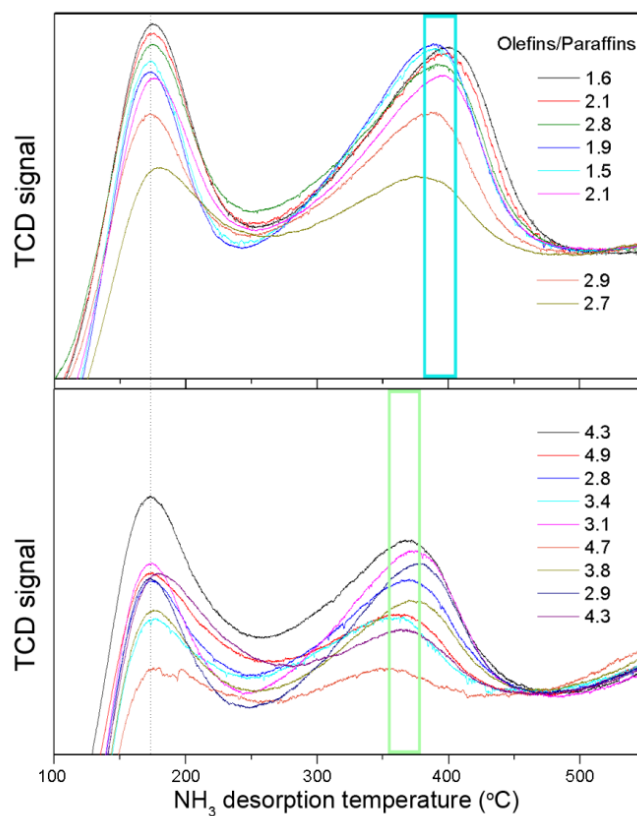


Fig. S16 NH_3 -TPD profiles of a series of MSAPO samples, which lead to products with varying selectivity ratios of olefins to paraffins ($\text{C}_2^= - \text{C}_4^=$)/($\text{C}_2^0 - \text{C}_4^0$). The selectivity ratios of olefins/paraffins are listed in the figure, demonstrating importance of medium strength acidity on the olefin selectivity. Reaction conditions: 400 $^{\circ}\text{C}$, 2.5 MPa and GHSV 5143 ml/h g_{cat} .

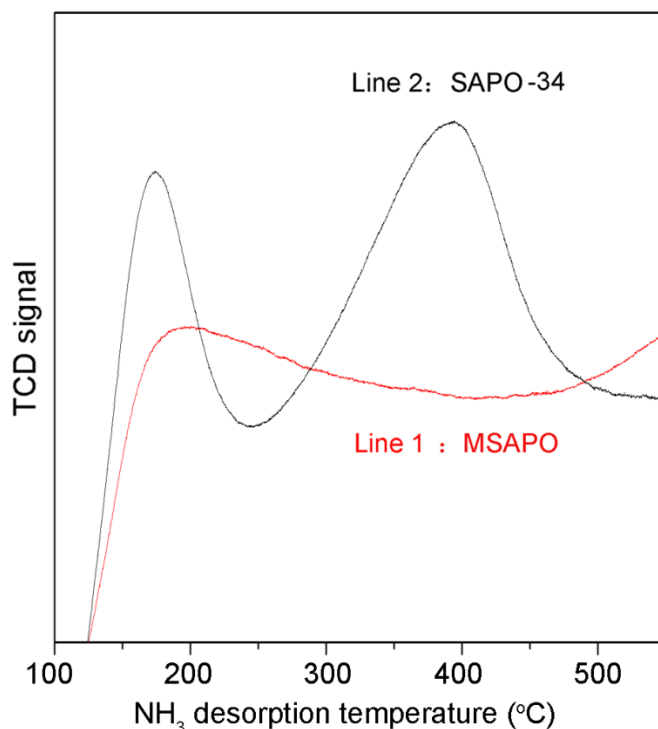


Fig. S17 NH₃-TPD profiles for a MSAPO (Line 1) and a commercially available microporous SAPO-34 (Line 2, purchased from Nankai University Catalyst Company, China). Line 1 represents a MSAPO with almost no medium strength acidic sites and the composite catalyst containing this MSAPO gives a similar product distribution as the ZnCrO_x only catalyst. Line 2 represents SAPO-34 with a peak maximum of medium acidity sites at ~394 °C, which gives a C₂^o–C₄^o paraffin selectivity over 50% and C₂⁼–C₄⁼ selectivity of 43% (Table S3). The results indicate that both the hierarchical pore structure and the medium strength acidity play important roles in determining the selectivity of olefins. By contrast, composite catalysts containing metal oxides (such as Cr₂O₃-ZnO, CuO-ZnO) and a variety of conventional zeolites (ZSM-5, Y, β) yielded mainly dimethyl ether (10), liquefied petroleum gas (C₃^o–C₄^o) (11-13) or gasoline (14, 15).

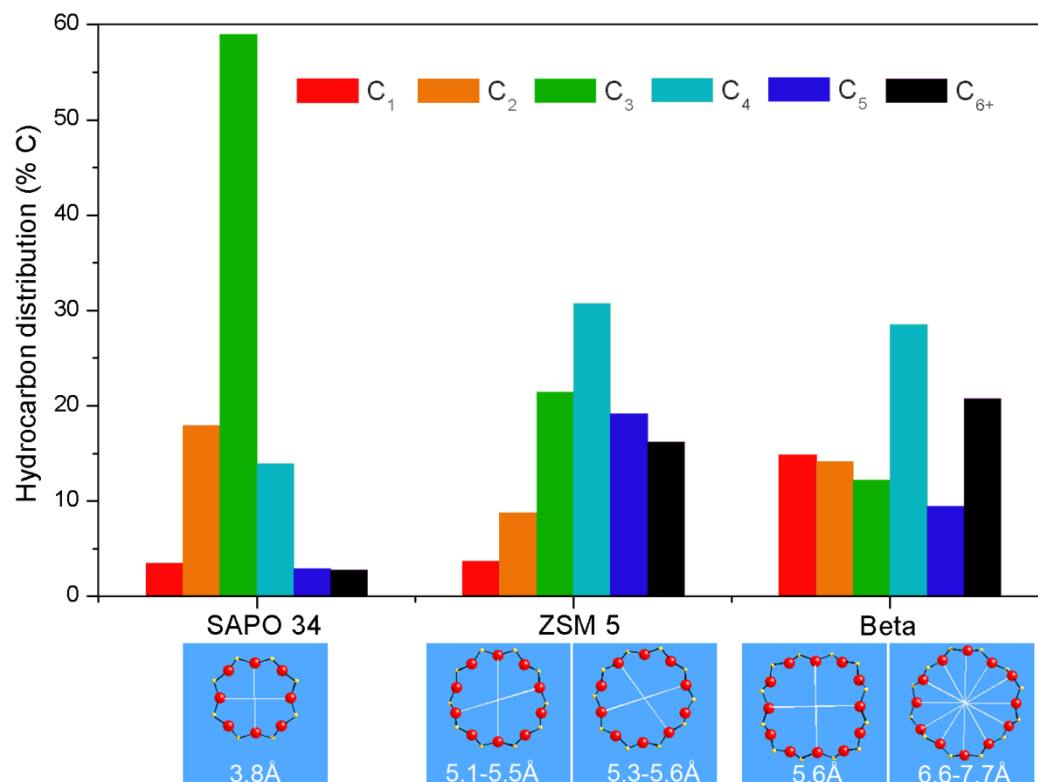


Fig. S18 Hydrocarbon distribution over composite catalysts containing different zeolites including the commercially available SAPO-34, ZSM-5 and Beta zeolites. Reaction conditions: 400 °C, 2.5 MPa and GHSV 5143 ml/h g_{cat} .

Table S1. Surface areas and pores of MSAPO and ZnCrO_x obtained from N₂ adsorption-desorption at 77 K.

Sample	BET specific surface area	Micro-area	Meso-area	Micropore volume	Pore volume	Average mesopore diameter
	m ² /g	m ² /g	m ² /g	cc/g	cc/g	nm
ZnCrO_x	98	0	98	0	0.28	~12
MSAPO	365	151	214	0.07	0.35	~7.6

Table S2. Detailed reaction results in Fig. 2A.

Space Velocity	CO Conversion %	CO ₂ Selectivity %	Methanol Selectivity %	Hydrocarbon distribution, %			
				CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅ +
Mode 1							
1285	7	34	1	53	19	19	9
3857	5	25	2	47	26	18	9
7714	4	29	2	48	30	13	9
Mode 2							
1285	9	37	a	26	23	46	5
3857	6	35	a	20	40	35	5
7714	4	48	a	16	54	26	4
Mode 3							
1285	25	41	a	8	45	42	5
3857	15	41	a	8	57	31	4
7714	10	42	a	7	65	24	4
Mode 4							
1285	35	40	a	3	69	20	8
3857	23	41	a	3	76	15	6
7714	17	41	a	2	80	14	4

a: the selectivity to methanol was below 0.5%C and was not reported in the list.

Table S3. Detailed reaction results of fig. S17.

	CO	CO₂	Methanol	Hydrocarbons distribution %			
	conversion	Selectivity	Selectivity	CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ^o -C ₄ ^o	C ₅₊
	%	%	%				
Line 1							
MSAPO	10	32	1	34	37	18	11
Line 2							
SAPO-34	32	39	a	4	43	50	3

a: the selectivity to methanol was below 0.5%C and was not reported in the list.

Table S4. The equilibrium concentration of methanol and ketene from thermodynamic calculations using HSC-9.0 software under 400 °C, 25 bar, with 1.5 kmol H₂ and 1 kmol CO as the starting compounds.

Components of the effluents	Methanol	ketene
Case 1: CO, H ₂ , CH ₃ OH (methanol)	1137 ppm	—
Case 2: CO, H ₂ , CH ₂ CO (ketene), CO ₂	—	8740 ppm
Case 3: CO, H ₂ , CH ₃ OH, CH ₂ CO, CO ₂	1089 ppm	8727 ppm
Case 4: CO, H ₂ , CH ₃ OH, CH ₂ CO, H ₂ O	1116 ppm	3267 ppm
Case 5: CO, H ₂ , CH ₃ OH, CH ₂ CO, H ₂ O, CO ₂	1085 ppm	9270 ppm

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